

Iterated-logarithmic pivot bounds for randomly pivoted Cholesky

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Abstract

We study the original coordinate version of randomly pivoted Cholesky: at each step a pivot is sampled in proportion to the diagonal of the current residual. For every finite complex Hermitian positive-semidefinite input, we prove expected trace error at most $(1 + \varepsilon)$ times the best rank- r trace error after

$$O(r/\varepsilon + r \log \log r) \quad \text{and, more sharply,} \quad O(r/\varepsilon + r \log \log \log \log r)$$

pivots. The logarithmic notation describes large-rank asymptotics; explicit formulas cover every rank and $0 < \varepsilon \leq 1$. The first proof combines a factorial likelihood comparison with a nonlinear determinant drift. The second retains an unordered fractional likelihood moment. Its central estimate combines an entropy inequality for the contributions accumulated along pivot orders with matrix transformations that preserve selected orders. A global maximization argument controls their correlation cost. An exact normalization transfers this estimate from an auxiliary pair to each terminal pivot set. Explicit finite-rank counts accompany both theorems. Julia experiments extend an existing kernel comparison and run the original pivot process for 100,000 pivots on three additional $131,072 \times 131,072$ kernel matrices. A two-sided multiplicative approximation to the pivot probabilities preserves the four-logarithm rank dependence, with explicit losses in the approximation factor. The rank-independent $O(r/\varepsilon)$ question remains open.

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1 Introduction and main results

1.1 The approximation problem

Kernel matrices can be large even when their significant spectral content is concentrated in a much smaller subspace. A useful approximation method must identify that subspace while inspecting only a small part of the matrix. Randomly pivoted Cholesky does this adaptively: each new column is chosen with probability proportional to the error remaining on its diagonal. The central theoretical question is how many such columns are needed to compete with the best rank- r approximation, uniformly over the spectrum and the coordinate representation.

Let $A \in \mathbb{C}^{n \times n}$ be Hermitian positive semidefinite, with eigenvalues $\lambda_1(A) \geq \dots \geq \lambda_n(A) \geq 0$, and define

$$\tau_r(A) = \sum_{j>r} \lambda_j(A), \quad \tau_0(A) = \text{tr } A.$$

The best rank- r approximation has trace-norm error $\tau_r(A)$. Randomly pivoted Cholesky starts at $R_0 = A$. If $T = \text{tr } R_k > 0$, it selects coordinate i with probability $(R_k)_{ii}/T$ and sets

$$R_{k+1} = R_k - \frac{R_k e_i e_i^* R_k}{(R_k)_{ii}}. \quad (1)$$

Coordinates with zero diagonal have zero probability. Once the residual is zero, it remains zero. The approximation is $\hat{A}_k = A - R_k$. The trace of R_k is both the total remaining diagonal error and the trace-norm approximation error. We seek a pivot count depending only on the target rank r and relative accuracy ε , such that $\mathbb{E} \text{tr } R_k \leq (1 + \varepsilon) \tau_r(A)$. In particular, the count should not deteriorate when the leading eigenvalues become arbitrarily large relative to the tail. All algorithmic expectations below concern the coordinate process (1).

Chen, Epperly, Tropp, and Webber [1] give the modern algorithmic and theoretical account of RPCholesky, including its connection with earlier adaptive sampling [4, 5]. Epperly [2] proves the expected-error guarantee with sufficient count

$$k \geq r/\varepsilon + 2r\sqrt{\log r} + r \log(1/\varepsilon) + 2.3r. \quad (2)$$

His conjectured $O(r/\varepsilon)$ count is the unrestricted question considered here. Divan and Gilles [3] study spectral-decay-dependent rates using elementary symmetric polynomials. Holden [7] proves explicit linear-count guarantees under specified tail envelopes and identifies limitations of some continuous-time comparisons. Our results impose no spectral-decay assumption and allow arbitrary complex eigenvectors.

The expressions $O(r \log \log r)$ and $O(r \log \log \log \log r)$ below describe the additional rank overhead at fixed accuracy. The full bounds always include r/ε . Neither statement proves an absolute $O(r/\varepsilon)$ count.

1.2 Main results

To state the rates without exceptions at small ranks, define $\Lambda_1(r) = \log(e + r)$ and $\Lambda_{j+1}(r) = \log(e + \Lambda_j(r))$. Thus Λ_2 and Λ_4 are positive versions of the twofold and fourfold iterated logarithms. All logarithms are natural.

Theorem 1.1 (Two-logarithm overhead). *For $r \geq 2$, let $K_2(r, \varepsilon)$ be the explicit count in (18), and set $K_2(1, \varepsilon) = \lceil 1/\varepsilon \rceil$. For every finite Hermitian PSD matrix $A \in \mathbb{C}^{n \times n}$, every $1 \leq r \leq n$, and $0 < \varepsilon \leq 1$, the original RPCholesky process satisfies*

$$\mathbb{E} \text{tr } R_k \leq (1 + \varepsilon) \tau_r(A) \quad \text{for every integer } k \geq \min\{n, K_2(r, \varepsilon)\}. \quad (3)$$

There is an absolute constant C_2 such that $K_2(r, \varepsilon) \leq C_2 r(\varepsilon^{-1} + \Lambda_2(r))$.

Theorem 1.2 (Four-logarithm overhead). *For $r \geq 2$, let $K_4(r, \varepsilon) = K_4(r, \varepsilon; 1/2)$ in (44), and set $K_4(1, \varepsilon) = \lceil 1/\varepsilon \rceil$. Under the same assumptions,*

$$\mathbb{E} \operatorname{tr} R_k \leq (1 + \varepsilon) \tau_r(A) \quad \text{for every integer } k \geq \min\{n, K_4(r, \varepsilon)\}. \quad (4)$$

There is an absolute constant C_4 such that $K_4(r, \varepsilon) \leq C_4 r(\varepsilon^{-1} + \Lambda_4(r))$. More generally, every $0 < \beta < 1$ gives a sufficient count $K_4(r, \varepsilon; \beta)$ through (44); this parameter can be optimized for finite ranks.

At fixed accuracy these are the asserted $O(r \log \log r)$ and $O(r \log \log \log \log r)$ bounds. Both theorems are uniform over finite dimensions and arbitrary complex eigenvectors. When $\tau_r(A) = 0$, the process exhausts the input in at most r positive pivots.

1.3 Why a determinant enters the proof

A direct trace argument explains the accuracy term r/ε but does not by itself give the desired uniform starting point. The exact one-step trace drift is

$$\mathbb{E}[\operatorname{tr} R_{k+1} \mid R_k] = \operatorname{tr} R_k - \frac{\operatorname{tr} R_k^2}{\operatorname{tr} R_k}.$$

Once the expected trace is a constant multiple of $\tau_r(A)$, this drift finishes the approximation in $O(r/\varepsilon)$ steps. The difficulty is to reach that regime without paying for the potentially enormous initial ratio $\operatorname{tr} A/\tau_r(A)$.

We use the determinant potential

$$P(R) = \det\left(I + \frac{r}{\tau_r(A)} R\right).$$

It serves two purposes. First, it controls the trace through $P(R) \geq 1 + (r/\tau_r(A)) \operatorname{tr} R$. Second, its minor expansion is compatible with volume sampling, which provides an estimate after the first r pivots that is independent of the initial spectral ratio. The determinant also has a stronger nonlinear drift:

$$\mathbb{E}[P(R_{k+1}) \mid R_k] \leq 2e P(R_k)^{1-1/r}.$$

Repeated application reduces the exponent of the initial potential. We refer to this phase as *cooling*.

Both proofs therefore have the same interface: establish $\mathbb{E} P(R_r)^\theta \leq e^H$ for some $0 < \theta \leq 1$. Cooling turns this into a controlled first moment; a stopped-potential estimate then brings the expected trace to a constant multiple of the tail; the trace drift supplies the final accuracy. For fixed θ , the cooling cost is $O(r[1 + \log(1 + H/r)])$. The whole improvement between the two theorems comes from reducing the scale of H :

Likelihood estimate	H/r	Additional rank overhead
Factorial comparison	$O(\log r)$	$O(r \log \log r)$
Fractional comparison	$O(\log \log \log r)$	$O(r \log \log \log \log r)$

1.4 Why the second proof uses occupation entropy

Let μ_r be the law of the unordered first r pivots and ν_r the size- r volume law. The elementary comparison $d\mu_r/d\nu_r \leq r!$ gives the first theorem. It treats all pivot orders by their largest possible cost. For the sharper theorem we retain the averaging over orders and terminal sets. If $L = d\mu_r/d\nu_r$, then Hölder's inequality gives

$$\mathbb{E}_{\mu_r} P^\theta \leq (\mathbb{E}_{\nu_r} L^{1/(1-\theta)})^{1-\theta} (\mathbb{E}_{\nu_r} P)^\theta.$$

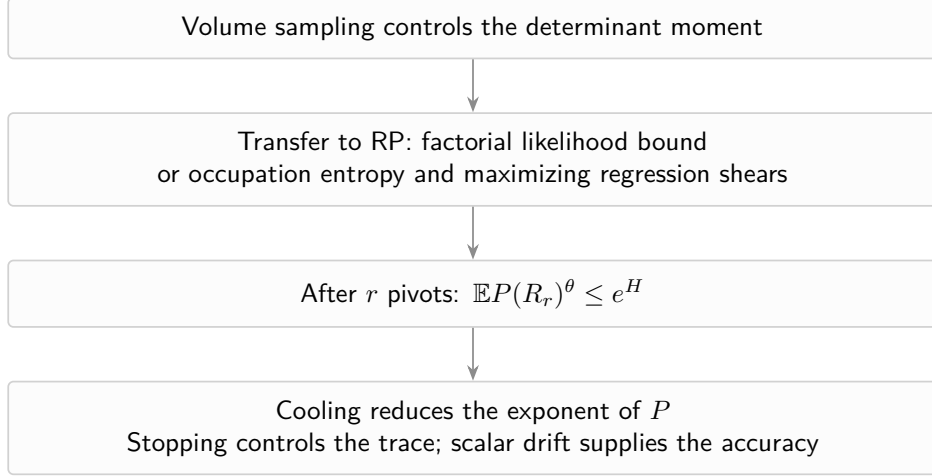


Figure 1: How the two arguments meet. Their different likelihood estimates produce different initial moment scales H ; the remaining conversion is the same. All algorithmic transitions use the original coordinate RP law.

Thus a fractional determinant moment can use an averaged likelihood estimate that would be lost in a worst-case comparison.

The difficult term is the likelihood norm. We encode its order sum in an auxiliary matrix partition. Differentiating that partition produces an *occupation matrix*, which records how much each direction contributes along an order. An entropy inequality relates the partition to the occupation matrix, leaving a mismatch between its diagonal and the covariance diagonal. At a joint maximum of a penalized partition, stationarity expresses this mismatch in terms of correlations. Regression shears on every coordinate subset then bound the total correlation contribution. Finally, an exact terminal normalization transfers this auxiliary estimate back to the original pivot law.

These steps are organized so that each estimate addresses the term left uncontrolled by the preceding one. Section 2 proves the common moment-to-error conversion; Section 3 completes the first theorem. Sections 4–6 establish the auxiliary likelihood estimate, and Section 7 transfers it and proves the second theorem. The finite counts and experiments follow in Sections 9 and 10.

The determinant-path comparison, trace drift, and volume-sampling identities have antecedents in [1, 2, 5]. We include proofs of the versions used here. The additional ingredients are the nonlinear moment conversion and the occupation, maximization, and normalization estimates that yield the four-logarithm overhead.

2 From a determinant moment to relative trace error

We first isolate the part of the argument shared by both theorems. The starting information will be a determinant moment after r pivots; this section explains how that information pays for a finite number of additional pivots. The three phases have separate roles: reduce the moment scale, enter a constant-error regime, and then attain the requested relative accuracy.

2.1 Residual order and the choice of potential

Write $e_j(R)$ for the j th elementary symmetric polynomial of the eigenvalues, with $e_0 = 1$ and $e_j = 0$ beyond the dimension. The identity $\det(I + tR) = \sum_j t^j e_j(R)$ also identifies $e_j(R)$ with the sum of the principal j -minors.

Lemma 2.1 (Order and exhaustion). *Every positive pivot satisfies $0 \preceq R_{k+1} \preceq R_k \preceq A$ and reduces the rank by one. Moreover,*

$$\text{tr } R_k \geq \tau_k(A), \quad \tau_r(R_k) \leq \tau_r(A).$$

Proof. For a pivot i , the vector $u = R^{1/2}e_i/\sqrt{R_{ii}}$ has norm one and lies in the range of R . Thus the update is $R^{1/2}(I - uu^*)R^{1/2}$. This proves order and rank reduction. Since $A - R_k$ is a PSD matrix of rank at most k bounded above by A , its trace is at most the sum of the first k eigenvalues of A . The other inequality follows from eigenvalue monotonicity. $\square$

Fix $\tau = \tau_r(A) > 0$. The preceding lemma shows that every residual still has rank- r tail at most τ . This suggests the scale $t = r/\tau$: the leading r eigenvalues each contribute at most $1/t$ to the logarithmic derivative of a determinant, while the remaining eigenvalues contribute at most their total mass τ . At this scale the two contributions are balanced. Set

$$P(R) = \det(I + tR), \quad t = r/\tau,$$

and write $p_R(z) = \det(I + zR)$ when differentiating in z . The trace identity below measures progress toward the final error target; the determinant identity is what makes the earlier cooling phase possible.

Lemma 2.2 (Exact determinant and trace drifts). *For a nonzero residual of trace T ,*

$$\mathbb{E}[P(R') \mid R] = \frac{p'_R(t)}{T}, \quad \mathbb{E}[\text{tr } R' \mid R] = T - \frac{\text{tr } R^2}{T}. \quad (5)$$

In particular,

$$\mathbb{E}[P(R') \mid R] \leq 2e P(R)^{1-1/r}. \quad (6)$$

The last inequality also holds at the absorbing zero residual.

Proof. The determinant lemma applied to (1), followed by averaging with probability R_{ii}/T , gives

$$\mathbb{E}[P(R') \mid R] = P(R) \left(1 - \frac{t}{T} \text{tr}[R^2(I + tR)^{-1}] \right) = \frac{P(R)}{T} \text{tr}[R(I + tR)^{-1}].$$

This is the first identity; the trace identity follows directly from the update. Zero diagonal entries have zero columns and lose no contribution.

Let $x\tau$ be the sum of the leading r eigenvalues of R , padding by zeros if necessary, and let $y\tau$ be the remaining trace. We call the leading r eigenvalues the *head* and the rest the *tail*. By Lemma 2.1, $0 \leq y \leq 1$. AM–GM on the head and the tail bound $\log(1 + u) \leq u$ imply

$$P(R)^{1/r} \leq (1 + x)e^y, \quad \frac{p'_R(t)}{P(R)} \leq \tau \left(\frac{x}{1 + x} + y \right).$$

The second inequality also uses concavity of $u/(1 + tu)$ on the leading r eigenvalues. Consequently

$$\frac{\mathbb{E}[P(R') \mid R]}{P(R)^{1-1/r}} \leq e^y \left(1 + \frac{xy}{x + y} \right) \leq 2e.$$

The last step uses $e^y \leq e$ and $xy/(x + y) \leq y \leq 1$. The case $R = 0$ has $P = 1$ and is immediate. $\square$

The determinant drift is useful far from the target. Near the target, the trace is the more direct quantity to follow. Its excess $u = \mathbb{E} \text{tr } R/\tau - 1$ decreases roughly like $u^2/[r(1 + u)]$. Integrating the reciprocal of this decrease suggests $1/u - \log u$ as a progress variable. The next lemma makes this calculus heuristic a discrete inequality.

Lemma 2.3 (Scalar completion). *Put $f_k = \mathbb{E} \operatorname{tr} R_k$. Then*

$$f_{k+1} \leq f_k - \frac{(f_k - \tau)_+^2}{r f_k}, \quad (7)$$

with the evident convention at zero. While $u_k = f_k/\tau - 1 > 0$, the function $F(u) = 1/u - \log u$ increases by at least $1/r$ per step, unless the desired threshold has already been reached.

Proof. The leading r eigenvalues have sum at least $(T - \tau)_+$, so $\operatorname{tr} R^2 \geq (T - \tau)_+^2/r$. The function $x \mapsto x - (x - \tau)_+^2/(rx)$ is nondecreasing and concave, as is checked separately on either side of τ . Apply Jensen to (5). For $u > 0$, the resulting comparison is $u' \leq u - u^2/[r(1 + u)]$. The decreasing convex function F satisfies

$$F\left(u - \frac{u^2}{r(1 + u)}\right) - F(u) \geq -F'(u) \frac{u^2}{r(1 + u)} = \frac{1}{r}.$$

Monotonicity of F gives the assertion for the actual means. $\square$

2.2 The three phases and their pivot counts

Suppose the first r pivots give the moment estimate $\mathbb{E}P(R_r)^\theta \leq e^H$. We now define the remaining counts in the order in which they are used. Put

$$c_d = \log(2e), \quad \zeta = 1 - 1/r.$$

After g cooling steps the exponent of the initial determinant is $\gamma = \zeta^g$. To use the known θ th moment we need $\gamma \leq \theta$. We also want the resulting log moment to be at most $r(c_d + \log 5)$. Solving these two requirements gives

$$g(r; \theta, H) = \left\lceil \frac{\log \max\{\theta^{-1}, [H/(r\theta) - c_d]/\log 5, 1\}}{-\log(1 - 1/r)} \right\rceil. \quad (8)$$

The maximum is at least one even if its middle entry is negative. We retain the actual moment estimate, rather than substituting its upper cap:

$$\gamma = (1 - 1/r)^g, \quad D(r; \theta, H) = (1 - \gamma)rc_d + \gamma H/\theta. \quad (9)$$

The number 5 has a concrete role. Above trace 10τ , the conditional mean of the next determinant is at most one fifth of its current value. Stopping its contribution when the trace crosses this threshold therefore converts a log moment D into a constant expected trace in

$$s(r; \theta, H) = \lceil D(r; \theta, H)/\log 5 \rceil \quad (10)$$

steps. The resulting normalized excess is at most $9 + 1/r$. Lemma 2.3 then requires only the increase $F(\varepsilon) - F(9 + 1/r)$, and gives the final count

$$n_{\text{acc}}(r, \varepsilon) = \left\lceil r \left[\varepsilon^{-1} + \log(\varepsilon^{-1}) + \log(9 + 1/r) - \frac{1}{9 + 1/r} \right] \right\rceil. \quad (11)$$

Thus each summand in

$$\mathcal{K}(r, \varepsilon; \theta, H) = r + g(r; \theta, H) + s(r; \theta, H) + n_{\text{acc}}(r, \varepsilon) \quad (12)$$

has a distinct purpose: initialization, cooling, stopping, and accuracy. Figure 2 draws the four phases to scale for one rank. The threshold 10τ is a fixed convenient choice, not an optimized parameter. The following proposition verifies all the stated phase estimates.

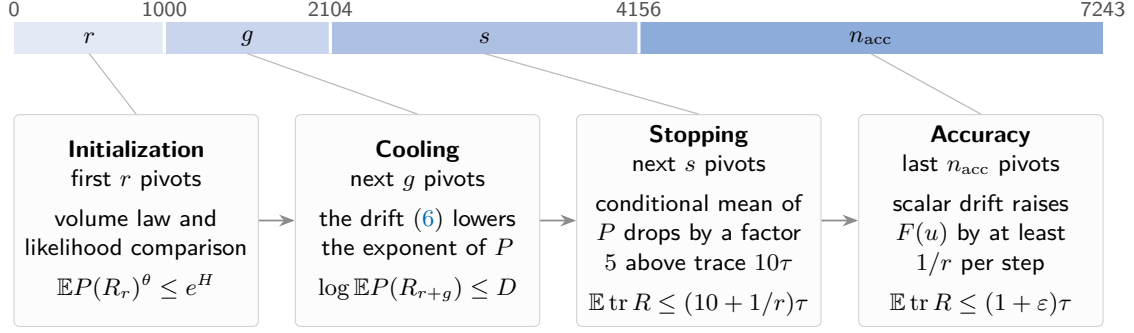


Figure 2: The four phases of the count (12). The bar is drawn to scale for the four-logarithm count at $r = 1000$ and $\varepsilon = 1$ in Table 1; tick labels are cumulative pivot numbers. Each box names the estimate used in its phase and the resulting guarantee. The count uses the choice of β in Section 9; the four phase lengths are 1000, 1104, 2052, and 3087. Since cooling stops once $D \leq r(c_d + \log 5)$, the stopping phase never needs more than $\lceil r(1 + c_d/\log 5) \rceil \approx 2.05r$ pivots.

Proposition 2.4 (Initial moment to explicit count). *Suppose $r \geq 2$ and*

$$\mathbb{E}P(R_r)^\theta \leq e^H, \quad 0 < \theta \leq 1. \quad (13)$$

Then the count (12) gives expected trace at most $(1 + \varepsilon)\tau$.

Proof. Let $\zeta = 1 - 1/r$. Conditional iteration of (6), using concavity at each step, yields

$$\mathbb{E}[P(R_{r+g}) \mid R_r] \leq (2e)^{r(1-\zeta^g)} P(R_r)^{\zeta^g}.$$

For g in (8), $\gamma = \zeta^g \leq \theta$ and $\gamma[H/(r\theta) - c_d] \leq \log 5$. The monotonicity of positive moments therefore gives

$$\log \mathbb{E}P(R_{r+g}) \leq (1 - \gamma)rc_d + \gamma H/\theta = D, \quad D \leq r(c_d + \log 5). \quad (14)$$

For every residual, head counting and the tail bound give

$$\frac{p'_R(t)}{P(R)} = \sum_j \frac{\lambda_j(R)}{1 + t\lambda_j(R)} \leq r/t + \tau = 2\tau.$$

Whenever $\text{tr } R > 10\tau$, this gives $\mathbb{E}[P(R') \mid R] \leq P(R)/5$. Because trace decreases pathwise, the stopped contribution $P(R_k)\mathbf{1}_{\{\text{tr } R_k > 10\tau\}}$ obeys the same conditional contraction: paths that have left the region cannot return. After $s = \lceil D/\log 5 \rceil$ steps, the expected surviving potential is at most one. Since $P(R) \geq t \text{tr } R$, we obtain

$$f_{r+g+s} \leq 10\tau + \tau/r.$$

Now $u \leq 9 + 1/r$. Lemma 2.3 and $F(\varepsilon) = \varepsilon^{-1} + \log \varepsilon^{-1}$ show that exactly the additional count (11) suffices. Early success only helps. If a specified count reaches n , exhaustion proves the capped conclusion. $\square$

Lemma 2.5 (Rank one). *For every PSD input, $\mathbb{E} \text{tr } R_k \leq (1 + 1/k)\tau_1(A)$ for $k \geq 1$.*

Proof. Write $\lambda = \lambda_1(A)$ and $\tau = \tau_1(A) > 0$. The first trace drift gives $f_1 \leq \lambda + \tau - \lambda^2/(\lambda + \tau) \leq 2\tau$. For $r = 1$, the scalar comparison sends $1 + 1/k$ to $1 + 1/(k + 1)$ after normalizing by τ . Induction proves the claim. Zero tails exhaust in one pivot. $\square$

3 Volume sampling and the two-logarithm theorem

We now supply the first initial moment. The determinant of a selected principal block appears naturally because the product of Cholesky pivot diagonals equals that determinant. Volume sampling assigns probability proportional to exactly the same quantity. It therefore separates the set-dependent determinant from the order-dependent trace denominators. This is the comparison measure for which the residual determinant moment is especially simple to calculate.

For a supported r -subset S , write $R_A(S)$ for the Schur residual after pivoting on S in any order, and set

$$\nu_r(S) = \frac{\det A_{SS}}{e_r(A)}.$$

This is the size- r volume distribution. Let μ_r denote the unordered set law of the first r original RP pivots. When $\tau_r(A) > 0$, these laws have the same positive-minor support.

Lemma 3.1 (A factorial comparison). *On that support, $\mu_r(S) \leq r! \nu_r(S)$.*

Proof. For an ordered pivot sequence with underlying set S , the product of the pivot diagonals is $\det A_{SS}$. Its probability is therefore

$$\frac{\det A_{SS}}{\prod_{j=0}^{r-1} \text{tr } R_j}. \quad (15)$$

The denominator is at least $\prod_{j=0}^{r-1} \tau_j(A)$. We also have $e_r(A) \leq \prod_{j=0}^{r-1} \tau_j(A)$. Indeed, for an ordered nonnegative list λ , separating its first entry gives

$$e_q(\lambda) = \lambda_1 e_{q-1}(\lambda_{2:}) + e_q(\lambda_{2:}) \leq \left(\sum_i \lambda_i \right) e_{q-1}(\lambda_{2:}),$$

where we used $e_q(\lambda_{2:}) \leq e_1(\lambda_{2:}) e_{q-1}(\lambda_{2:})$. Indeed, in the product on the right each degree- q square-free monomial occurs q times, and all other terms are nonnegative. Iteration proves the product bound. Summing (15) over the at most $r!$ orders of S proves the lemma. This is the determinant-path mechanism used in the preceding RP analyses [5, 2]. $\square$

The factorial comparison alone is useful only if the volume law has a small determinant moment. Its advantage is algebraic: a selected minor times a residual minor is a larger minor of the original matrix. Summing over selected sets counts each larger minor a known number of times. Only the spectral tail remains to be estimated.

For the resulting absolute constant write

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad c_v = 1 + 2 \log \varphi + \varphi^{-1} < 2.581. \quad (16)$$

The golden ratio enters solely through minimizing the elementary one-variable bound at the end of the next proof.

Lemma 3.2 (Volume determinant moment). *For $\tau = \tau_r(A) > 0$ and $P(S) = \det(I + (r/\tau)R_A(S))$,*

$$\mathbb{E}_{\nu_r} P(S) \leq e^r L_r(-r) \leq e^{c_v r}, \quad L_r(-x) := \sum_{j=0}^r \binom{r}{j} \frac{x^j}{j!}. \quad (17)$$

Proof. Schur determinant factorization and minor counting give

$$\mathbb{E}_{\nu_r} e_j(R_A(S)) = \binom{r+j}{j} \frac{e_{r+j}(A)}{e_r(A)}.$$

Each degree- $(r+j)$ eigenvalue monomial contains at least j tail variables. Thus $e_{r+j}(A) \leq e_r(A)e_j(\lambda_{r+1:}) \leq e_r(A)\tau^j/j!$. It follows that

$$\mathbb{E}_{\nu_r} P \leq \sum_{j \geq 0} \binom{r+j}{j} \frac{r^j}{j!} = e^r L_r(-r).$$

The identity follows by multiplying e^x by the finite sum defining $L_r(-x)$ and using Vandermonde's identity coefficient by coefficient. For any $u > 0$, the elementary bound $(ru)^j/j! \leq e^{ru}$ gives

$$L_r(-r) \leq e^{ru} \sum_{j=0}^r \binom{r}{j} u^{-j} = e^{ru} (1 + 1/u)^r.$$

The minimizing u is φ^{-1} , and $1 + \varphi = \varphi^2$. This proves (17). $\square$

Combining the two comparisons gives a concrete choice of initial data for the conversion proposition:

$$H_r^{(2)} = \log(r!) + c_v r, \quad K_2(r, \varepsilon) = \mathcal{K}(r, \varepsilon; 1, H_r^{(2)}), \quad r \geq 2. \quad (18)$$

This formula is valid at every finite rank; the asymptotic statement follows by inspecting its cooling term.

Proof of Theorem 1.1. Lemmas 3.1 and 3.2 give $\mathbb{E}P(R_r) \leq r!e^{c_v r}$. Apply Proposition 2.4 with $\theta = 1$ and $H = H_r^{(2)}$. Since $\log(r!)/r = O(\log r)$, $g = O(r \log \log r)$, while $s = O(r)$ and $n_{\text{acc}} = O(r/\varepsilon)$. Lemma 2.5 and exhaustion handle the remaining cases. Replacing the asymptotic iterated logarithm by $\Lambda_2(r)$ absorbs the finitely many small ranks into an absolute constant. $\square$

4 An auxiliary pair and its occupation matrices

The factorial loss arises from estimating every order separately. To reduce it, we need to study the whole order sum before bounding it. Inverse principal matrices provide a convenient way to encode that sum: their selected diagonal entries describe successive determinant ratios, while their traces against a second positive matrix describe the normalizing denominators. This leads to a finite pair $K, W \succ 0$ of order r .

For interpretation, K describes the geometry of a full-rank selected core and W measures its residual trace. In Section 7 we will construct the precise pair associated with each terminal set of the original input. For now allowing arbitrary positive pairs lets us prove a uniform inequality. Principal submatrices are denoted by K_R, W_R ; embedding a matrix on R means adding zero rows and columns outside R . Gradients on Hermitian matrices use the real trace inner product.

For a nonempty coordinate set $R \subseteq [r]$, define

$$U_R = K_R^{-1}, \quad T_R = \text{tr}(W_R U_R), \quad \Gamma_R = \text{embed}_R(U_R/T_R).$$

Define the positive partition recursively by

$$Z_\emptyset = 1, \quad Z_R = \sum_{i \in R} \frac{(U_R)_{ii}}{T_R} Z_{R-i}. \quad (19)$$

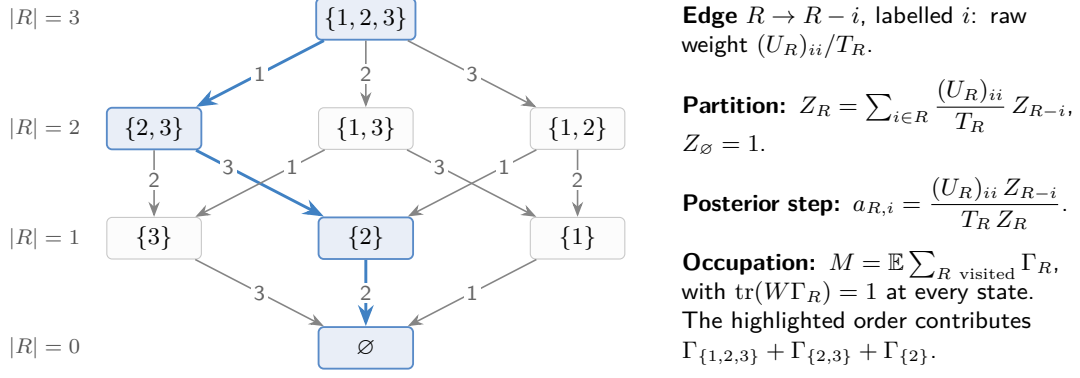


Figure 3: The order sum defining $Z_{[r]}$ for $r = 3$. A state is the set R of coordinates not yet deleted; an edge labelled i deletes i . Z_R is the total raw weight of all paths from R to $\emptyset$. Weighting each edge by its child partition and dividing by the parent partition gives the posterior order law (21), and M averages Γ_R over the states a posterior path visits. Highlighted: the order 1, 3, 2.

We use

$$\begin{aligned} \mathcal{L}(K, W) &= \det(W) Z_{[r]}, & C &= W^{-1}, \\ Q(W) &= \det(W) \prod_i C_{ii} = \det \text{corr}(C)^{-1} \geq 1. \end{aligned} \quad (20)$$

The factor $\det W$ makes $\mathcal{L}$ invariant under simultaneous diagonal changes of scale. The quantity Q measures the departure of the covariance C from a diagonal matrix, where $\text{corr}(C)_{ij} = C_{ij}/\sqrt{C_{ii}C_{jj}}$. It equals one for diagonal C and grows when the normalized covariance determinant becomes small. We will prove an envelope of the form $\mathcal{L} \leq e^B Q^\beta$. The power of Q keeps track of the correlation cost that will later cancel against terminal sampling weights.

An order has raw weight $\prod_R (U_R)_{ii}/T_R$; dividing by $Z_{[r]}$ gives a probability distribution on orders, which we call the posterior order law. Its actual next-label probabilities are

$$a_{R,i} = \frac{(U_R)_{ii} Z_{R-i}}{T_R Z_R}. \quad (21)$$

In particular, the child normalizer is part of the law. Figure 3 shows the recursion, the posterior law, and the occupation matrix defined next for $r = 3$.

To control the order sum, perturb its trace denominators. The logarithmic derivative records the average cumulative contribution of the visited states. Accordingly, let M be the expected sum of Γ_R along a posterior order, and put $\mu_i = M_{ii}$. We call M the occupation matrix. Since each visited state contributes one to $\text{tr}(W\Gamma_R)$,

$$\text{tr}(WM) = r, \quad -\nabla_W \log Z_{[r]} = M. \quad (22)$$

The derivative follows by expanding the finite partition over complete orders and differentiating their linear trace denominators. Every order visits the full set first, so $M \succeq \Gamma_{[r]} = K^{-1}/\text{tr}(WK^{-1}) \succ 0$. Four scalar quantities separate the contributions we need to estimate. The determinant deficit J measures the nonuniformity of the eigenvalues of $W^{1/2}MW^{1/2}$, whose sum is r . The correlation deficit C_M compares the product of the occupation diagonals with its determinant. The information quantity I compares the same product with the partition. The quantity S_1 measures the occupation diagonals relative to the covariance

diagonals. Explicitly,

$$\begin{aligned} I &= \sum_i \log \mu_i - \log Z_{[r]}, & J &= -\log \det(WM), \\ C_M &= \sum_i \log \mu_i - \log \det M, & S_1 &= \sum_i \log(\mu_i/C_{ii}), \quad q = \log Q. \end{aligned} \tag{23}$$

Then $J \geq 0$ by AM–GM and (22), and direct cancellation gives

$$\log \mathcal{L} = C_M - I - J, \quad C_M = S_1 + q + J. \tag{24}$$

The quantity I will be nonnegative by its explicit recursion below; no unstated differential-entropy assertion is needed.

5 Entropy interpolation

The identities above express $\log \mathcal{L}$ as a correlation gain minus two deficits. We seek an inequality that lets us exchange part of the correlation gain for information. The free parameter α measures that exchange. Its eventual choice will leave a coefficient of q no larger than the exponent β in the desired envelope. The one term that this argument cannot bound on its own is S_1 ; controlling it is the task of the maximization argument in the next section.

For probability vectors a, v on the same finite set, write $D_{\text{KL}}(a\|v) = \sum_i a_i \log(a_i/v_i)$ and $H(a) = -\sum_i a_i \log a_i$, with the usual zero-mass convention. We use only these finite discrete entropies.

Proposition 5.1 (Occupation–determinant interpolation). *For every positive pair and $0 < \alpha < 1$,*

$$\log \mathcal{L} \leq \alpha(S_1 + q) - (1 - \alpha)J - (r - 1) \log \alpha. \tag{25}$$

Proof. The partition is recursive, so we compare each state with the posterior-weighted average of its actual children. The information increment will be a sum of relative entropies. The determinant increment will retain a nonnegative concavity reserve; discarding that reserve would lose the interpolation we need.

At a state of dimension $d \geq 2$, omit the state subscript. Write $U = K^{-1}$, $T = \text{tr}(WU)$, $a_i = (U_{ii}/T)Z_{-i}/Z$, and let M_i be the occupation mean of the actual child, embedded with zero i th row and column when necessary. Then

$$N_i = U/T + M_i, \quad M = \sum_i a_i N_i.$$

Define $x_i = U_{ii}/(T\mu_i)$ and $\chi = \sum_i x_i$. We have $0 < x_i < 1$ and $0 < \chi < d$. For each coordinate j , define a probability vector

$$\vartheta_j^{(j)} = x_j, \quad \vartheta_i^{(j)} = a_i(M_i)_{jj}/\mu_j \quad (i \neq j).$$

The identity $M = U/T + \sum_i a_i M_i$ proves normalization. Substituting $a_i = (U_{ii}/T)Z_{-i}/Z$ into (23) gives

$$D_0 := I - \sum_i a_i I_i = \sum_j D_{\text{KL}}(a\|\vartheta^{(j)}) \geq 0. \tag{26}$$

Child determinants and information quantities in these formulas are taken on the $(d - 1)$ -dimensional child space.

The determinant does not obey an exact additive chain rule under mixtures. We keep its surplus over the corresponding child expression as the determinant reserve

$$R_0 = \log \det M - \sum_i a_i \log \det M_i - \sum_i a_i \log(U_{ii}/T).$$

The Schur complement of U at coordinate i is K_{-i}^{-1} , so

$$\det N_i = (U_{ii}/T) \det M_i \det(I + M_i^{-1} K_{-i}^{-1}/T).$$

It follows that $R_0 \geq \Delta_{\det} := \log \det M - \sum_i a_i \log \det N_i \geq 0$, where the last sign is concavity of log determinant. Algebra also gives

$$C_M - \sum_i a_i C_{M_i} = D_0 + H(a) - R_0, \quad H(a) = - \sum_i a_i \log a_i. \quad (27)$$

The next two estimates use the same scalar $\chi = \sum_i U_{ii}/(T\mu_i)$. When the information increment is small, χ is forced close to one; that restriction also bounds the remaining entropy gain. This shared dependence is the reason for retaining the determinant reserve. Let $X_i = M^{-1/2} N_i M^{-1/2}$ and $v_i = M^{1/2} e_i / \sqrt{\mu_i}$. Then $\|v_i\| = 1$, $v_i^* X_i v_i = x_i$, and $\sum_i a_i X_i = I$. Taking a Schur complement in the v_i direction and using $\log \det S \leq \text{tr } S - (d-1)$ shows

$$\log \det X_i \leq \log x_i + \text{tr } X_i - x_i - (d-1).$$

Average to obtain $\Delta_{\det} \geq \sum_i a_i (x_i - 1 - \log x_i)$. Consequently

$$H(a) - R_0 \leq 1 + \log \chi - D_{\text{KL}}(a \| x/\chi) - \sum_i a_i x_i \leq 1 + \log \chi. \quad (28)$$

On the other hand, grouping the j th atom and its complement in each term of (26), followed by the log-sum inequality, gives

$$D_0 \geq -\log \chi + (d-1) \log \frac{d-1}{d-\chi} \geq \chi - 1 - \log \chi. \quad (29)$$

The final inequality is $-\log(1-u) \geq u$ with $u = (\chi-1)/(d-1) < 1$; it remains valid for negative u .

For any $\eta > 0$, (28)–(29) imply

$$H(a) - R_0 - \eta D_0 \leq 1 + \log \chi - \eta(\chi - 1 - \log \chi) \leq (1 + \eta) \log(1 + 1/\eta).$$

The maximum occurs at $\chi = (1 + \eta)/\eta$. In dimension one, $I = C_M = 0$. At every state, the preceding inequalities give

$$C_M - \sum_i a_i C_{M_i} \leq (1 + \eta) \left(I - \sum_i a_i I_i \right) + (1 + \eta) \log(1 + 1/\eta).$$

Average this inequality at each depth under the posterior order law, then sum over depths. The a_i are the conditional child probabilities, so the tower property makes both differences telescope from the root to the one-dimensional states. There are exactly $r-1$ such increments. In particular (26) also gives $I \geq 0$ at the root. We obtain

$$C_M \leq (1 + \eta)I + (r-1)(1 + \eta) \log(1 + 1/\eta).$$

With $\alpha = \eta/(1 + \eta)$, this becomes $I \geq (1 - \alpha)C_M + (r-1) \log \alpha$. Combining it with (24) proves the proposition. $\square$

6 Joint maximization and hereditary shears

For a general pair, the occupation diagonal need not be close to the covariance diagonal, so the entropy estimate leaves S_1 uncontrolled. An envelope for all pairs can instead be proved at the largest value of $\log \mathcal{L} - \beta \log Q$. At such a maximum, variation in W ties the occupation matrix to the covariance. Variation of the whole pair then supplies restrictions on that covariance.

There are two issues to resolve. A maximizing sequence may send K to a singular boundary, and transformations of different child states cannot be chosen independently because all states originate from one matrix. The compactness argument below preserves the complete family of principal inverses from a single sequence. The subsequent regression shears are common transformations of the original pair. These choices allow both stationarity and comparisons between whole order events.

Lemma 6.1 (Existence of a maximizing limit). *Fix r and $0 < \beta < 1$. The supremum of $\mathcal{L}(K, W)/Q(W)^\beta$ over positive pairs is attained in a limit of actual pairs, at a finite positive W . The limiting partition is smooth in every positive test precision matrix. At a maximizing limit its occupation mean satisfies*

$$M = (1 - \beta)C + \beta C \operatorname{diag}(1/C_{ii})C, \quad C = W^{-1}. \quad (30)$$

All limiting order probabilities come from one common sequence of positive pairs.

Proof. First $\mathcal{L} \leq r!$. At a remaining set R with next coordinate i , put $c_i = (W_R^{-1})_{ii}$ and $b_i = (K_R^{-1})_{ii}$. Since $W_R \succeq e_i e_i^*/c_i$, one has $T_R \geq b_i/c_i$. Along a complete order, $\prod_R c_i = 1/\det W$, so its contribution to $\det WZ$ is at most one. There are $r!$ orders. For diagonal W , induction in (19) gives $Z_R = 1/\prod_{i \in R} W_{ii}$ and hence $\mathcal{L} = Q = 1$. The supremum therefore lies in $[1, r!]$.

Simultaneous congruence $(K, W) \mapsto (DKD, DWD)$ by a positive diagonal D preserves $\mathcal{L}$ and Q . Choose $D_{ii} = \sqrt{C_{ii}}$, so that the new inverse precision is $\Omega = \operatorname{corr} C$. A sequence with score at least $\delta > 0$ has $Q \leq (r!/\delta)^{1/\beta}$. Thus $\operatorname{diag} \Omega = 1$, $\operatorname{tr} \Omega = r$, and $\det \Omega$ is bounded below. These conditions place Ω , and hence $W = \Omega^{-1}$, in a compact positive region.

There is a simple compactification that avoids classifying spectral degenerations. For each nonempty R normalize the principal inverse by

$$G_R^{(m)} = (K_R^{(m)})^{-1} / \operatorname{tr}((K_R^{(m)})^{-1}).$$

Each $G_R^{(m)}$ is PSD of trace one. There are finitely many sets R ; extract one subsequence along which all $G_R^{(m)}$ converge and the normalized precisions converge to $W_0 \succ 0$. In the partition recursion replace U_R/T_R by $G_R/\operatorname{tr}(W_R G_R)$, which is unchanged before taking the limit. For test precisions $aI \preceq W \preceq bI$,

$$b^{-|R|} \leq Z_R(W) \leq a^{-|R|}, \quad 0 \preceq G_R/\operatorname{tr}(W_R G_R) \preceq I/a.$$

The first bounds follow inductively because $\operatorname{tr} G_R = 1$. Consequently the finite recursions, their first derivatives, and all order probabilities converge, uniformly on compact positive precision sets. Zero limiting transitions contribute zero; their child partitions stay bounded. This closure retains compatibility by extraction from a single $K^{(m)}$ sequence. It does not admit independently chosen state matrices.

Apply this construction to a maximizing sequence. For every fixed test precision $V \succ 0$, its limiting score is bounded by the same supremum, because each $(K^{(m)}, V)$ is an admissible positive pair. The limiting partition therefore attains its global precision maximum at

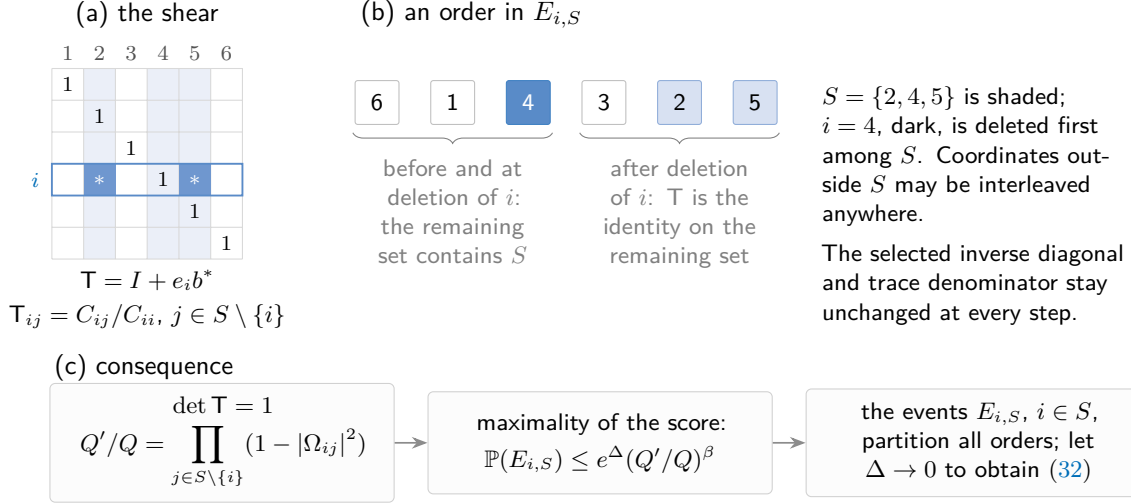


Figure 4: The hereditary shear argument for one subset S and one $i \in S$. (a) The shear changes only row i , in the columns of S . (b) On every order whose first deletion from S is i , all raw path weights are unchanged. (c) The shear lowers Q by the regression factor, so global maximality bounds the probability of that order event; summing over $i \in S$ and letting the score gap Δ tend to zero gives (32). Here $\Omega = \text{corr } C$.

W_0 . Differentiating its finite smooth recursion gives $\nabla \log Z = -M$. The gradient of $(1 - \beta) \log \det W + \log Z - \beta \sum_i \log C_{ii}$ is zero there, which is exactly (30). In particular $M \succ 0$. The interpolation inequality passes to this limit by continuity of its matrix means, determinants, and positive diagonal entries. $\square$

Stationarity reduces the desired mean bound to a sum of squared covariance correlations. To control those correlations, regress a chosen coordinate out of selected others. This reduces their marginal variances and hence decreases Q . At the same time it preserves every pivot order in which the chosen coordinate is first among that subset. Maximality forces the probability of this preserved event to be no larger than the corresponding correlation penalty. Applying the argument to every subset, rather than only to the whole coordinate set, permits successive removal of low-degree vertices. Figure 4 summarizes the construction.

For later use abbreviate

$$b_r = \log \left(1 + \frac{2 \log(r!)}{r} \right). \quad (31)$$

Lemma 6.2 (Hereditary first-coordinate constraint). *At a joint maximizing limit as above, let $\Omega = \text{corr } C$. For every nonempty $S \subseteq [r]$,*

$$\sum_{i \in S} \prod_{j \in S \setminus \{i\}} (1 - |\Omega_{ij}|^2)^\beta \geq 1. \quad (32)$$

Consequently

$$S_1 \leq r \log \left(1 + \frac{2 \log(r!)}{r} \right) = r b_r. \quad (33)$$

Proof. We first work with a finite positive pair. Fix $i \in S$, and form the unit-diagonal shear $T = I + e_i b^*$ whose only nonzero off-diagonal entries are

$$T_{ij} = C_{ij}/C_{ii}, \quad j \in S \setminus \{i\}.$$

Transform both matrices by $(K, W) \mapsto (TKT^*, TWT^*)$. On any complete deletion order in which i is first among S , every raw path weight is unchanged. To see this, before deleting

i , no changed row uses an already deleted column. At each remaining set the congruence is exactly T_R ; the trace $\text{tr}(W_R K_R^{-1})$ and the selected inverse diagonal are preserved. After deleting i , the shear is the identity on the remaining coordinates. All coordinates outside S may be interleaved arbitrarily.

The new covariance is $C' = \mathsf{T}^{-*} C \mathsf{T}^{-1}$. Its diagonal satisfies $C'_{jj} = C_{jj} - |C_{ij}|^2 / C_{ii}$ for $j \in S \setminus \{i\}$, with all other diagonal entries unchanged. Since $\det \mathsf{T} = 1$,

$$Q'/Q = \prod_{j \in S \setminus \{i\}} (1 - |\Omega_{ij}|^2). \quad (34)$$

Let $E_{i,S}$ denote the preserved order event and let Δ be the difference between the global supremum of the log score and the current log score. The transformed partition includes all old weights on $E_{i,S}$, so its total is at least $Z\mathbb{P}(E_{i,S})$. Global comparison therefore implies

$$\mathbb{P}(E_{i,S}) \leq e^\Delta (Q'/Q)^\beta.$$

Along the common maximizing sequence, $\Delta \rightarrow 0$. The covariance, order probabilities, and the right side of (34) converge. For a fixed S , the events $E_{i,S}$ partition all orders. Summing their limiting probabilities proves (32). This argument uses global competitors at the root, not independently maximizing children.

Put $w_{ij} = -\log(1 - |\Omega_{ij}|^2) \geq |\Omega_{ij}|^2$. For a subset of size m , (32) ensures a vertex with internal weighted degree at most $\log m / \beta$. Remove one such vertex successively. Every edge is counted once, giving

$$\sum_{i < j} w_{ij} \leq \frac{\log(r!)}{\beta}, \quad \sum_i \sum_{j \neq i} |\Omega_{ij}|^2 \leq \frac{2 \log(r!)}{\beta}.$$

Taking diagonal entries of (30) yields $\mu_i / C_{ii} = 1 + \beta \sum_{j \neq i} |\Omega_{ij}|^2$. Concavity of logarithm now proves (33). $\square$

The mean bound closes the entropy estimate at the joint maximum. Its remaining parameter balances a linear cost $rb_r \alpha$ against a logarithmic cost $-(r-1) \log \alpha$. For $r \geq 2$, define

$$\alpha_{r,\beta} = \min \left\{ \beta, \frac{r-1}{rb_r} \right\}, \quad B_{r,\beta} = rb_r \alpha_{r,\beta} - (r-1) \log \alpha_{r,\beta}. \quad (35)$$

For $r = 1$, set $\alpha_{1,\beta} = \beta$ and $B_{1,\beta} = 0$ directly; the quotient in (35) is not used. The constraint $\alpha \leq \beta$ ensures that the unused correlation term has a nonpositive coefficient.

Theorem 6.3 (Fractional envelope). *For all positive $r \times r$ pairs and every $0 < \beta < 1$,*

$$\mathcal{L}(K, W) \leq e^{B_{r,\beta}} Q(W)^\beta. \quad (36)$$

For $r = 1$ the assertion is exact with $B_{1,\beta} = 0$. For fixed β , $B_{r,\beta} = O(r \log \log \log r)$.

Proof. At the attained joint maximum, use Proposition 5.1 and Lemma 6.2. For $0 < \alpha \leq \beta$,

$$\log \mathcal{L} - \beta q \leq \alpha r b_r - (r-1) \log \alpha,$$

because $(\alpha - \beta)q$ and $-(1 - \alpha)J$ are nonpositive. Minimizing over α gives exactly (35). Bounding the global supremum bounds every finite pair. Finally $b_r = \log \log r + \log 2 + o(1)$ and, for fixed β , $B_{r,\beta}/r = 1 + \log b_r + o(1)$. $\square$

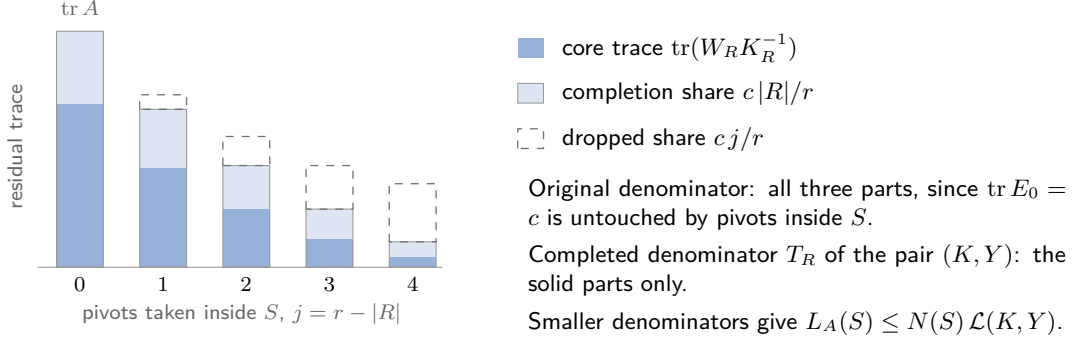


Figure 5: Trace denominators along one order inside a terminal set S , with illustrative values for $r = 5$. After j pivots the original residual trace is the core trace plus the outside trace $c = \text{tr } E_0$. The completion $Y = W + (c/r)K$ charges only $c|R|/r$ of the outside trace: the solid bars show the completed denominators, while their dashed extensions recover the original ones. This is a likelihood comparison; the original algorithm retains the whole outside residual. At $j = 0$ both denominators equal $\text{tr } A$.

7 Exact completion normalization and the terminal moment

The auxiliary envelope is not yet an estimate for the original algorithm: selecting r coordinates from a larger matrix leaves a residual outside the selected core. We must account for that residual in every earlier trace denominator. If that residual has trace c , the auxiliary construction charges c/r for each core coordinate that remains. The resulting denominator is no larger than the original one at every stage, so it gives an upper comparison of likelihoods.

This completion introduces a terminal-dependent normalizer N . Keeping N together with the volume probability and the correlation factor Q is essential. A set with an unfavorable covariance geometry may be sufficiently rare under volume sampling to compensate for that geometry. The identity proved below preserves precisely this compensation. Figure 5 illustrates the denominator comparison.

Fix a supported terminal S of size r for the original input A , and put

$$\begin{aligned}
 A_S &= A_{SS}, & K &= A_S^{-1}, & X &= A_{:S} A_S^{-1}, \\
 A_c &= X A_S X^*, & E_0 &= A - A_c \succeq 0, & c &= \text{tr } E_0, \\
 W &= X^* X, & Y &= W + (c/r)K, & V &= Y^{-1}, \\
 N(S) &= \frac{e_r(A) \det K}{\det Y}, & Q(S) &= \det Y \prod_{i \in S} V_{ii}.
 \end{aligned} \tag{37}$$

Here $X_S = I$ and E_0 has zero rows and columns on S . The matrix A_c is the rank- r interpolant supported by the selected columns. The added term $(c/r)K$ contributes $c|R|/r$ to a remaining core trace, which explains its coefficient. Write

$$a_r = r^r / r!.$$

Lemma 7.1 (Terminal comparison and weighted normalization). *Let $L_A(S) = \mu_r(S) / \nu_r(S)$. Then*

$$L_A(S) \leq N(S) \mathcal{L}(K, Y), \quad 0 < N(S) \leq a_r = r^r / r!, \tag{38}$$

and, for every $p \geq 1$,

$$\mathbb{E}_{\nu_r}[N^p Q] \leq 2^r a_r^p. \tag{39}$$

Proof. For an order using only coordinates in S , let $R \subseteq S$ denote those not yet pivoted. The remaining original trace is $c + \text{tr}(W_R K_R^{-1})$. The completed trace is $c|R|/r + \text{tr}(W_R K_R^{-1})$, no larger. Summing reciprocal trace products in (15) gives the first inequality in (38); the core normalization is $\mathcal{L}(K, Y) = (\det Y / \det K) \sum_{\omega} \prod_R T_R^{-1}$.

The elementary coefficient inequality for PSD sums is $e_r(A_c + E_0) \leq \sum_{j=0}^r e_{r-j}(A_c) e_j(E_0)$. For completeness, factor $A_c = FF^*$ and $E_0 = GG^*$ and put $\mathcal{G} = [F \ G]^* [F \ G]$. The matrices $\mathcal{G}$ and $A_c + E_0$ have the same nonzero eigenvalues, hence the same e_r . A principal r -minor of $\mathcal{G}$ selects $r - j$ columns of F , indexed by I , and j columns of G , indexed by J . The Hadamard–Fischer inequality for this PSD block gives

$$\det \begin{pmatrix} F_I^* F_I & F_I^* G_J \\ G_J^* F_I & G_J^* G_J \end{pmatrix} \leq \det(F_I^* F_I) \det(G_J^* G_J).$$

Sum over I, J and j . The two sums of principal minors are $e_{r-j}(F^* F) = e_{r-j}(A_c)$ and $e_j(G^* G) = e_j(E_0)$, which proves the coefficient inequality, including singular blocks. Since $e_j(E_0) \leq c^j/j!$,

$$e_r(A) \leq \sum_{j=0}^r e_{r-j}(A_c) c^j/j!, \quad \frac{\det Y}{\det K} = \sum_{j=0}^r e_{r-j}(A_c) (c/r)^j.$$

For $0 \leq j \leq r$, $r^j/j! \leq r^r/r!$. Comparing these nonnegative sums proves the normalizer bound.

A pointwise estimate of Q would undo the advantage of averaging over terminal sets. Instead, the factors introduced above cancel terminalwise:

$$\nu_r(S) N(S) Q(S) = \prod_{i \in S} V_{ii}. \quad (40)$$

To sum the remaining product over sets, we seek one common coordinate bound for all terminal matrices V . Ridge leverage scores provide such a bound because every terminal residual trace is at least the optimal tail. Suppose first rank $A > r$. Let $\xi_0 = \tau_r(A)/r > 0$ and $\omega_i = [A(A + \xi_0 I)^{-1}]_{ii}$. Then $\sum_i \omega_i \leq 2r$. For $\xi = c/r \geq \xi_0$, direct substitution gives

$$V = A_S [(A^2)_{SS} + \xi A_S]^{-1} A_S \preceq [A(A + \xi I)^{-1}]_{SS} \preceq [A(A + \xi_0 I)^{-1}]_{SS}.$$

For the first order, put $Z = A^{1/2} E_S$, where E_S selects the columns in S , and use $Z[Z^*(A + \xi I)Z]^{-1} Z^* \preceq (A + \xi I)^{-1}$; this is the orthogonal projection inequality after congruence by $(A + \xi I)^{1/2}$. Thus $V_{ii} \leq \omega_i$. Summing (40) over supported sets gives

$$\mathbb{E}_{\nu_r} NQ \leq e_r(\omega) \leq (2r)^r/r! = 2^r a_r.$$

At rank $A = r$, $E_0 = 0$ and $N = 1$; V is the selected block of the global range projection. Its common diagonal marks sum to r , which is even stronger. Finally $N^{p-1} \leq a_r^{p-1}$ proves (39). $\square$

We can now see the role of the fractional exponent. Raising $\mathcal{L} \leq e^{B_{r,\beta}} Q^\beta$ to the power $1/\beta$ leaves exactly one power of Q , which is the quantity controlled by the weighted normalization. Hölder then pairs the resulting likelihood norm with the first volume determinant moment. The complete log moment is

$$H_{r,\beta}^{(4)} = B_{r,\beta} + \log a_r + \beta r \log 2 + (1 - \beta) c_v r. \quad (41)$$

Here the four summands come respectively from the auxiliary envelope, completion normalization, ridge-score summation, and volume moment.

Proposition 7.2 (Unordered fractional moment). *For $0 < \beta < 1$, $p = 1/\beta$, and $\theta = 1 - \beta$,*

$$\mathbb{E}_{\nu_r} L_A^p \leq e^{pB_{r,\beta}} 2^r a_r^p, \quad (42)$$

$$\mathbb{E} P(R_r)^\theta \leq \exp(H_{r,\beta}^{(4)}). \quad (43)$$

Proof. Apply the pointwise envelope (36) to each terminal's own (K, Y) . Since $p\beta = 1$, Lemma 7.1 gives

$$\mathbb{E}_{\nu_r} L_A^p \leq e^{pB_{r,\beta}} \mathbb{E}_{\nu_r} N^p Q \leq e^{pB_{r,\beta}} 2^r a_r^p.$$

Hölder's inequality with conjugate exponents $1/\beta$ and $1/\theta$ then gives

$$\mathbb{E}_{\mu_r} P^\theta = \mathbb{E}_{\nu_r} L_A P^\theta \leq (\mathbb{E}_{\nu_r} L_A^{1/\beta})^\beta (\mathbb{E}_{\nu_r} P)^\theta.$$

Use (17) and (42) to obtain (41). □

For every $0 < \beta < 1$ this yields the explicit sufficient count

$$K_4(r, \varepsilon; \beta) = \mathcal{K}(r, \varepsilon; 1 - \beta, H_{r,\beta}^{(4)}), \quad r \geq 2. \quad (44)$$

Each choice is valid; taking the smallest of several such counts preserves the guarantee.

Proof of Theorem 1.2. Apply Proposition 2.4 to (43). For fixed β , the envelope theorem gives $H_{r,\beta}^{(4)}/r = O(\log \log \log r)$, because $\log a_r \leq r$. Consequently $g = O(r \log \log \log \log r)$, $s = O(r)$, and $n_{\text{acc}} = O(r/\varepsilon)$. Taking, for example, $\beta = 1/2$ gives an absolute constant in the theorem. The positive function $\Lambda_4(r)$ absorbs the finitely many small ranks. Lemma 2.5 supplies the stated count for $r = 1$; Lemma 2.1 handles zero tails and exhaustion. In both theorems the guarantee persists for every larger count, because the residual trace decreases along each sample path. □

8 Robustness to approximate pivot probabilities

An implementation may only approximate the residual diagonal probabilities. The determinant phases need an upper comparison of sampling probabilities, whereas the final trace phase needs a lower comparison. Keeping these roles separate gives an explicit robustness statement. Approximate pivoting itself is established prior work; see Divan and Gilles [3]. The result here transfers an initial determinant moment through all four phases.

At every nonzero residual let $\pi_i = R_{ii}/\text{tr } R$. Suppose that, conditional on the complete past, the implemented probabilities satisfy

$$a\pi_i \leq q_i \leq a^{-1}\pi_i, \quad 0 < a \leq 1. \quad (45)$$

The Schur updates remain exact. In particular zero diagonal entries are never chosen, and the zero residual is absorbing. No independence between the probability errors at different steps is assumed.

Proposition 8.1 (Explicit robust conversion). *Let $r \geq 2$, $\tau = \tau_r(A) > 0$, $t = r/\tau$, and $P(R) = \det(I + tR)$. Suppose the exact RP law satisfies $\mathbb{E}_\pi P(R_r)^\theta \leq e^H$, where $0 < \theta \leq 1$. For $0 < \varepsilon \leq 1$ put*

$$\begin{aligned} H_a &= H + r \log(1/a), & c_a &= \log(2e/a), \\ g_a &= \left\lceil \frac{\log \max\{\theta^{-1}, [H_a/(r\theta) - c_a]/\log 5, 1\}}{-\log(1 - 1/r)} \right\rceil, \end{aligned} \quad (46)$$

$$\begin{aligned} \rho_a &= (1 - 1/r)^{g_a}, & D_a &= (1 - \rho_a)rc_a + \rho_a H_a/\theta, \\ s_a &= \lceil D_a/\log 5 \rceil, & u_a &= 10/a - 1 + 1/r, \\ F(u) &= u^{-1} - \log u, & n_a &= \left\lceil \frac{r}{a} (F(\varepsilon) - F(u_a)) \right\rceil. \end{aligned} \quad (47)$$

Then $K_a = r + g_a + s_a + n_a$ pivots under (45) suffice for $\mathbb{E}_q \text{tr } R_{K_a} \leq (1 + \varepsilon)\tau$. The count may be capped at n .

Proof. An ordered path of length r has likelihood ratio at most a^{-r} relative to exact RP. Consequently its nonnegative terminal potential obeys $\mathbb{E}_q P(R_r)^\theta \leq e^{H_a}$. Next, the upper probability comparison and the exact determinant identity give, for $T = \text{tr } R > 0$,

$$\mathbb{E}_q[P(R') \mid \text{past}] \leq \frac{p'_R(t)}{aT} \leq \frac{2e}{a} P(R)^{1-1/r}, \quad \frac{p'_R(t)}{P(R)} \leq 2\tau. \quad (48)$$

Here $p_R(z) = \det(I + zR)$. To recall why the bounds are uniform, let $x\tau$ and $y\tau$ be the head and tail traces of R at rank r . Schur monotonicity gives $0 \leq y \leq 1$. AM–GM and concavity give $P^{1/r} \leq (1+x)e^y$ and $p'_R/P \leq \tau[x/(1+x) + y]$; their product after division by T is at most $2e$. Also $p'_R/P \leq r/t + \tau = 2\tau$. The zero residual satisfies the nonlinear bound directly.

Conditional iteration and concavity now give $\log \mathbb{E}_q P(R_{r+g_a}) \leq D_a \leq r(c_a + \log 5)$. The choice of g_a ensures both $\rho_a \leq \theta$ and $\rho_a[H_a/(r\theta) - c_a] \leq \log 5$. Above trace $10\tau/a$, the first bound in (48) contracts the determinant by a factor at most $1/5$. Trace decreases on each path, so the determinant multiplied by the indicator of remaining above this threshold also contracts by $1/5$. After s_a further steps its expectation is at most one. Since $P(R) \geq t \text{tr } R$, the resulting mean trace is at most $10\tau/a + \tau/r$.

Finally the lower comparison, applied to the nonnegative trace decrease, gives

$$\mathbb{E}_q[\text{tr } R' \mid \text{past}] \leq T - a \frac{\text{tr } R^2}{T} \leq T - \frac{a(T - \tau)_+^2}{rT}.$$

The last expression is an increasing concave function of T , so the same inequality holds for consecutive mean traces. For their normalized excess $u > 0$, the decrease is at least $au^2/[r(1+u)]$. The decreasing convex function F therefore increases by at least a/r per step until the target is reached. This proves the stated n_a . $\square$

For rank one, upper domination of the remaining trace after the first pivot gives $\mathbb{E}_q \text{tr } R_1 \leq 2\tau_1(A)/a$. The same scalar argument shows that

$$1 + \left\lceil a^{-1}[F(\varepsilon) - F(2/a - 1)] \right\rceil$$

pivots suffice, again capped at n . If $\tau_r(A) = 0$, positive pivots exhaust the input in at most r steps.

Taking $\theta = 1 - \beta$ and $H = H_{r,\beta}^{(4)}$ from (41), with fixed $0 < \beta < 1$, gives

$$K_a = O\left(\frac{r}{a\varepsilon} + \frac{r}{a} \log \frac{e}{a} + r\Lambda_4(r)\right).$$

Thus a fixed multiplicative probability error preserves the four-logarithm rank dependence. At $a = 1$ the finite formulas recover the original conversion exactly. This corollary uses the initial moment proved above; it does not replace that proof or validate an approximate Schur update.

9 Evaluating the explicit guarantees

The asymptotic improvement concerns very slowly varying functions, so the finite formulas are needed for numerical comparisons. Equations (18) and (44) use the same conversion; their difference is the initial moment. In the four-logarithm formula the exponent β balances the strength of the likelihood estimate against the smaller determinant exponent $1 - \beta$.

The effective initial scale in cooling is $H_{r,\beta}^{(4)}/(1 - \beta)$. Minimizing it reduces to one scalar equation. The minimizer lies in the branch $\alpha_{r,\beta} = \beta$ and is the unique root

$$(r - 1)(\beta^{-1} - 1 + \log \beta) = r(b_r + \log 2) + \log a_r, \quad 0 < \beta < 1. \quad (49)$$

Indeed, differentiate

$$c_v r + \frac{r b_r \beta - (r - 1) \log \beta + \log a_r + \beta r \log 2}{1 - \beta}.$$

Its derivative has the sign of the right side of (49) minus the left side. The left side decreases continuously from infinity to zero. Beyond $\beta = (r - 1)/(r b_r)$, when this boundary is below one, B is constant and the objective is increasing. At the boundary the derivative from the first branch is also positive, so the root precedes it. This proves the stated characterization. We use this choice throughout the experiments. It minimizes the moment scale, not necessarily the final count with all integer ceilings.

Target rank r	Two-logarithm count	Four-logarithm count
10	66	71
100	718	722
1,000	7,580	7,243
10,000	78,704	72,539
100,000	809,588	726,105

Table 1: Sufficient pivot counts at $\varepsilon = 1$ from (18) and (44). Counts are displayed before capping by the matrix dimension. The smaller asymptotic overhead need not give the smaller count at a small target rank.

To turn a pivot-count guarantee into an error curve, fix the available budget k and vary the comparison rank and accuracy. A small comparison rank permits a better accuracy parameter but has a larger spectral tail; a large rank has the opposite tradeoff.

For a known spectrum, each plotted theoretical error is

$$\min \left\{ 1, \min_{\substack{r \in \{1, \dots, n-1\}, \\ K(r, \varepsilon) \leq k}} (1 + \varepsilon) \frac{\tau_r(A)}{\text{tr } A} \right\}. \quad (50)$$

Our two curves use their respective theorems separately, both with the sharp rank-one option. They are not the minimum with a prior published bound. For fixed r, k , the smallest feasible ε is optimal. Writing $x = 1/\varepsilon$, we solve the monotone equation $x + \log x = a$ for $x \geq 1$ and

retain the feasible side of the numerical bracket. Since the remaining budget is integral, this inverts the final ceiling directly.

The prior curves use Known Result 1.1 and Corollary 1.3 of [2], including all their displayed constants. The former has sufficient count

$$r/\varepsilon + r \log(1/\varepsilon) + \min\{-r \log \eta_r, (\log 2)r^2 + r\}, \quad \eta_r = \tau_r(A)/\text{tr } A.$$

The latter uses (2). Their $\varepsilon = 1$ endpoint is interpreted by continuity. The curves optimize over integer r , not over a coarse accuracy grid.

10 Kernel experiments

We compare the explicit guarantees at finite budgets and measure their remaining gap to empirical RPCholesky error. Each theorem is evaluated with its own optimized comparison rank and accuracy. The figures use consistent colors and line styles; lower panels resolve differences between theoretical bounds that nearly overlap on logarithmic error axes.

10.1 Reproducing and extending the published comparison

Figures 6 and 7 reproduce the two kernel settings of Figure 1 in Epperly [2] and add the two guarantees proved here. The input points form an inclusive 80×80 grid on $[0, 1]^2$, so $n = 6400$. With bandwidth $\ell = 0.3$, we use the conventions

$$k_{\text{SE}}(x, y) = e^{-\|x-y\|^2/(2\ell^2)}, \quad k_{5/2}(x, y) = (1 + z + z^2/3)e^{-z}, \quad z = \sqrt{5}\|x - y\|/\ell.$$

If A_0 denotes the unshifted kernel matrix, the experimental input is $A = A_0 + 10^{-14}\lambda_{\max}(A_0)I$. The source specifies the bandwidth but does not give an explicit kernel formula; the formulas above fix the convention used in this reproduction. The empirical curves are means of our own ten independent trials.

The squared-exponential spectrum is the Kronecker product of two one-dimensional spectra. For Matérn-5/2, reflection in the two grid axes gives four invariant subspaces of dimension 1600; diagonalizing the corresponding blocks recovers the complete spectrum. All 6400 eigenvalues, including the shift, are retained. Tail sums are accumulated from the smallest eigenvalue upward. The color assignment follows the source figure’s legend: orange is Known Result 1.1 and pink is Corollary 1.3. The caption of the source figure interchanges those two color descriptions.

10.2 Larger inputs and additional kernels

To study budgets through 100,000, we use three additional $n = 131,072$ point inputs on the grid $x_i = i/(n + 1)$:

$$\begin{aligned} k_{\text{OU}}(x, y) &= e^{-|x-y|/0.3}, \\ k_{\text{BM}}(x, y) &= \min(x, y), \\ k_{\text{BB}}(x, y) &= \min(x, y) - xy. \end{aligned} \tag{51}$$

The exponential kernel is also the one-dimensional Laplace or Matérn-1/2 kernel. Laplace kernels occur in RPCholesky kernel regression experiments [1]. Brownian motion and Brownian bridge provide nonstationary covariance examples, complementing the stationary kernels. RPCholesky also has applications to kernel quadrature and Sobolev kernels [6]. No shift or spectral truncation is used for the three large inputs.

The Gaussian Markov property permits exact interval conditioning. Sampling an interval in proportion to its residual trace, then a coordinate in proportion to its conditional variance, gives exactly the probability in (1). Appendix A derives the variances, spectra, and implementation. Figures 8–10 show ten independent trials per kernel through 100,000 pivots.

10.3 Results

Table 2 reports endpoint errors and Monte Carlo standard errors. At 100,000 pivots, the four-logarithm guarantee is about 0.815 times Corollary 1.3 for each large kernel, while remaining about 13.6 times the empirical mean. For Matérn-5/2 the corresponding ratios are 0.708 and about 110. These examples illustrate both the finite improvement and the conservatism of a uniform guarantee.

The three large examples have closely related spectral tails. The discrete formulas in Appendix A give j^{-2} decay for indices large compared with the inverse length scale but small compared with n . At higher indices their spectra retain essentially the same discrete sine profile, up to scale and boundary corrections. Thus these experiments test chiefly one tail shape, with different boundary conditions and stationarity properties. When the cap at one is inactive, multiplying the normalized tail curve by a constant multiplies both spectrum-independent guarantees in (50) by that constant, leaving their ratio unchanged. This explains the close theoretical ratios, rather than supplying three independent tests of spectral-decay diversity.

The empirical agreement has a further geometric explanation. On a short interval of length dh , expansion of (53) gives

$$v_j^{\text{OU}} = \frac{2}{0.3} \frac{h j(d-j)}{d} (1 + O(dh/0.3)).$$

Apart from this common factor, the local bridge variances agree with the Brownian formula (52); Brownian motion and bridge differ only at the outer boundary. This similarity is consistent with their close late-budget RP errors. It is not a conclusion from eigenvalue decay alone, since coordinate RP also depends on eigenvectors. The four-logarithm bound divided by the empirical mean is respectively 13.6407, 13.6499, and 13.6490 for the exponential, Brownian-motion, and Brownian-bridge inputs.

For the squared-exponential input, the diagonal shift dominates the late-budget spectral tail, and our two endpoint bounds agree to the displayed precision. The early-budget ratios in the lower panel need not vary smoothly: the optimizing comparison rank and the sufficient integer count change discretely.

Kernel	k	Mean	Std. error	Two-log	Four-log
Squared-exp.	4000	8.522×10^{-12}	2.08×10^{-15}	2.269×10^{-11}	2.269×10^{-11}
Matérn-5/2	4000	1.322×10^{-7}	3.60×10^{-10}	1.587×10^{-5}	1.455×10^{-5}
Exponential	100000	6.606×10^{-6}	3.38×10^{-9}	9.638×10^{-5}	9.011×10^{-5}
Brownian motion	100000	1.980×10^{-6}	5.25×10^{-10}	2.891×10^{-5}	2.703×10^{-5}
Brownian bridge	100000	5.941×10^{-6}	2.31×10^{-9}	8.674×10^{-5}	8.109×10^{-5}

Table 2: Relative trace errors at the endpoint of each experiment. Means and standard errors use ten independent RPCholesky trials. The last two columns are the separate sufficient guarantees, evaluated using (50).

Squared-exponential kernel

80 × 80 grid · 10 independent trials · 4,000 pivots

— Known Result 1.1 — Two-log theorem — RPCholesky mean
- - Corollary 1.3 — Four-log theorem ····· Best rank- k

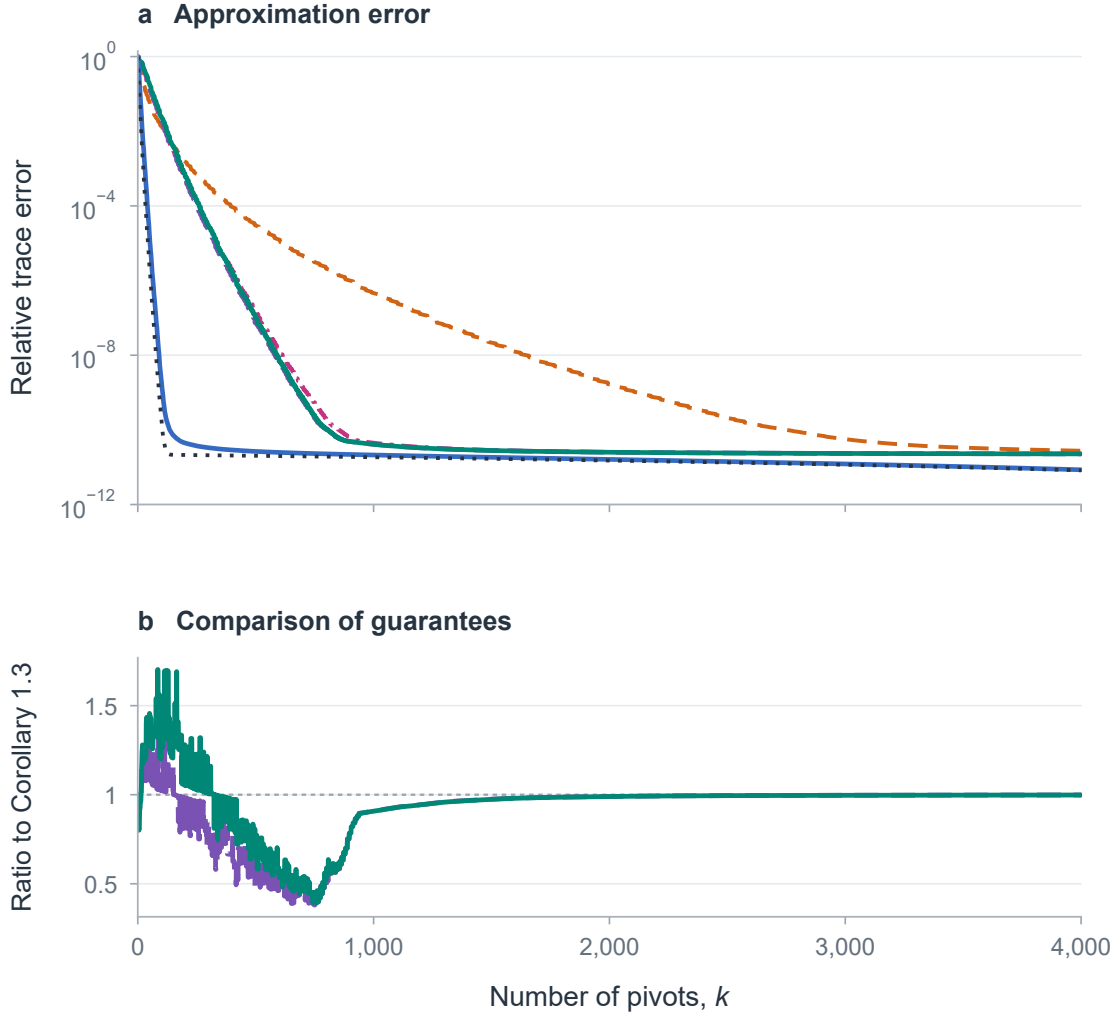


Figure 6: Squared-exponential kernel on the 80×80 grid. This vector redrawing of the comparison in Epperly [2], Figure 1, includes our two theorem bounds and our own ten-trial empirical mean. The upper panel shows relative trace error. The lower panel divides each new guarantee by Corollary 1.3; values below one indicate a smaller sufficient error bound. The late-budget spectral tail is dominated by the diagonal shift.

Matérn-5/2 kernel

80×80 grid · 10 independent trials · 4,000 pivots

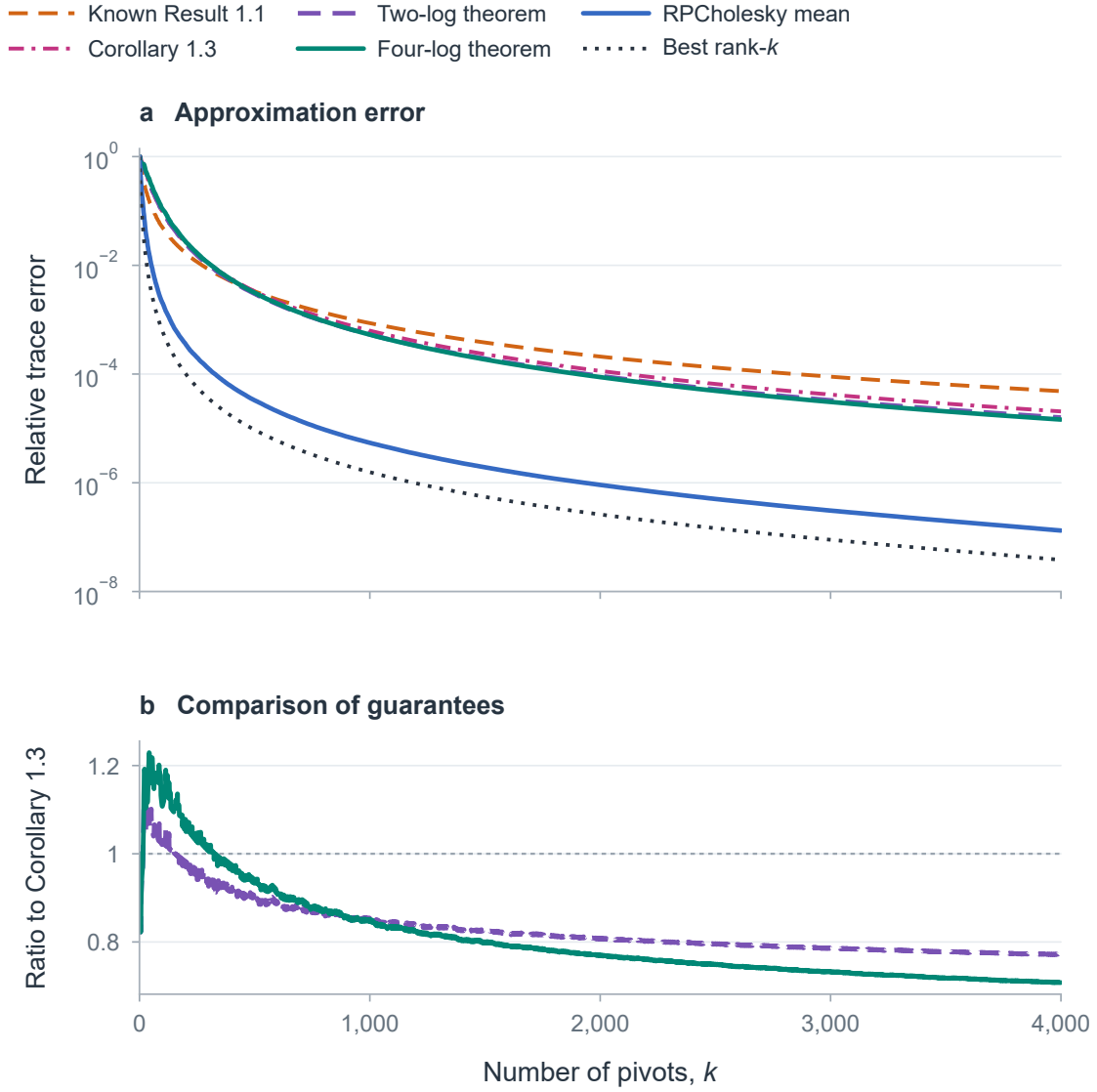


Figure 7: Matérn-5/2 kernel on the 80×80 grid, reproducing and extending the second comparison in Epperly [2], Figure 1. The empirical curve is the mean of ten independent trials. Both new bounds improve on Corollary 1.3 over much of the displayed range; the lower panel resolves the differences between the nearly overlapping upper curves.

Exponential / Matérn-1/2 kernel

131,072 points · 10 independent trials · 100,000 pivots

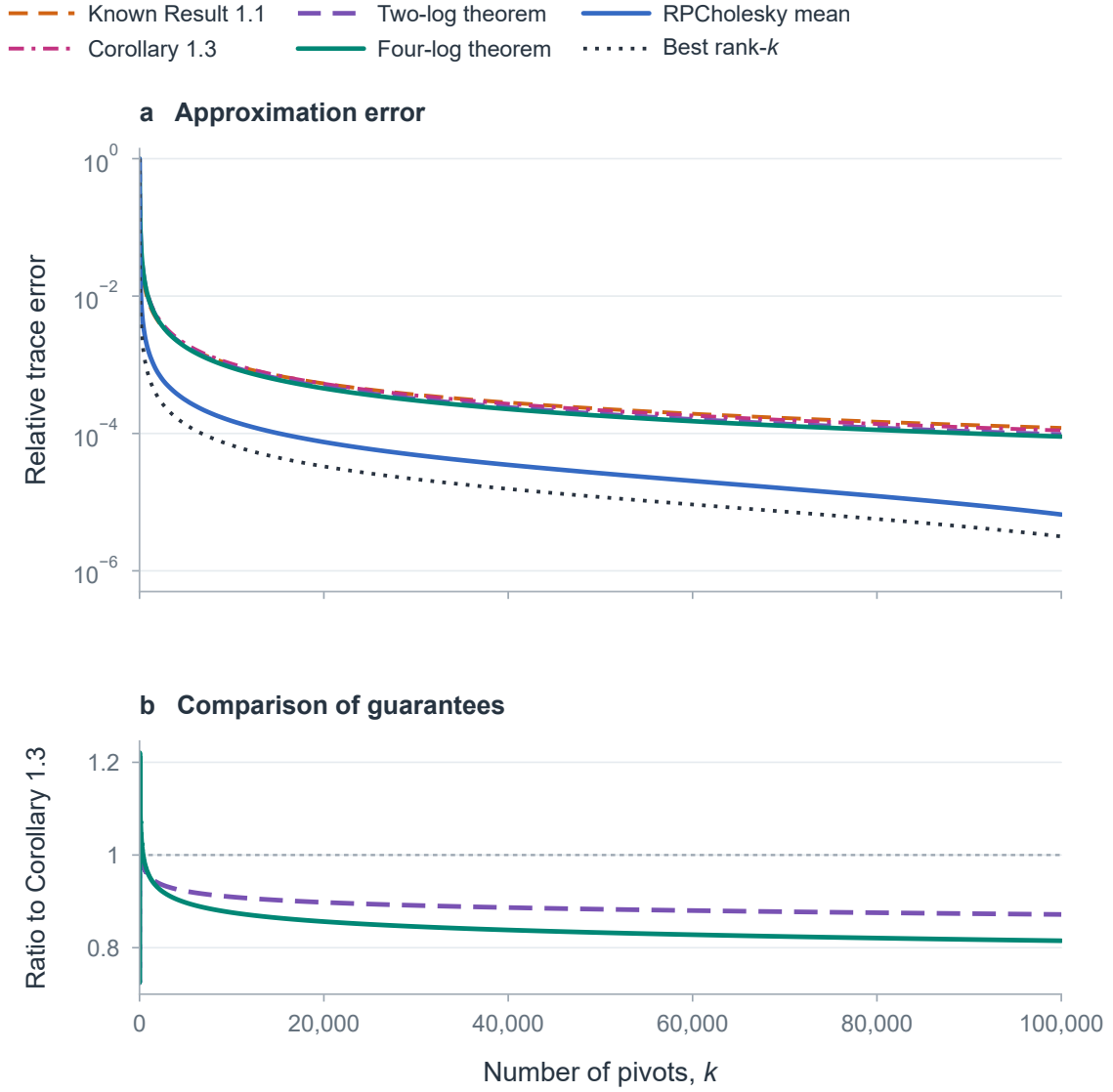


Figure 8: Exponential (Matérn-1/2) kernel with bandwidth 0.3 on 131,072 equally spaced interior points of $[0, 1]$. Ten original-RP trials are run through 100,000 pivots using exact interval conditioning. The black curve uses the complete discrete spectrum. At the endpoint, the four-logarithm guarantee is 9.011×10^{-5} , compared with empirical mean 6.606×10^{-6} .

Brownian motion kernel

131,072 points · 10 independent trials · 100,000 pivots

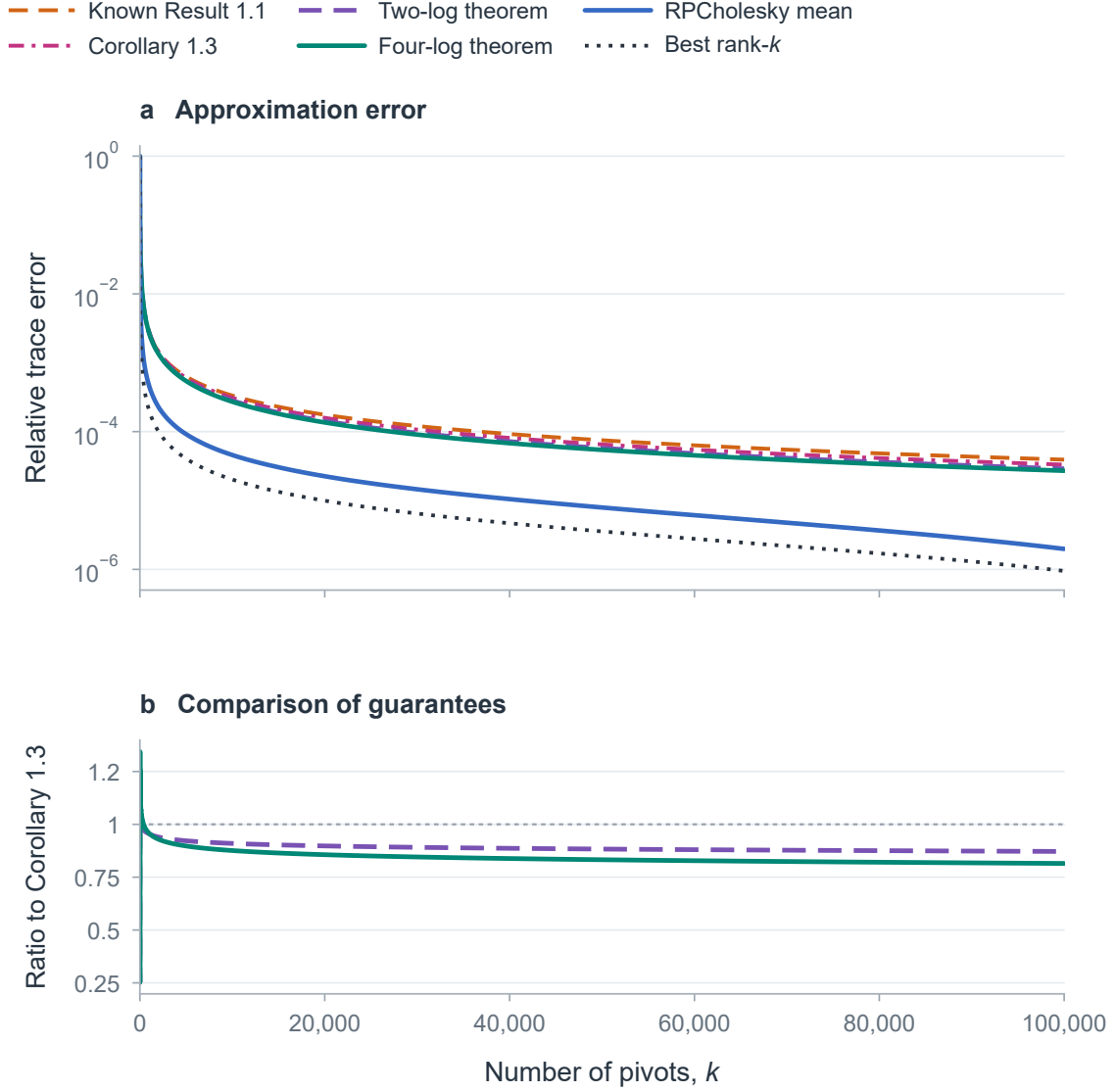


Figure 9: Brownian motion kernel $\min(x, y)$ on 131,072 equally spaced interior points. The empirical curve averages ten original-RP trials through 100,000 pivots. The upper panel compares the errors and the complete-spectrum optimum; the lower panel compares the two new guarantees with Corollary 1.3. The four-logarithm endpoint guarantee is 2.703×10^{-5} , and the empirical mean is 1.980×10^{-6} .

Brownian bridge kernel

131,072 points · 10 independent trials · 100,000 pivots

-- Known Result 1.1 -- Two-log theorem — RPCholesky mean
-.- Corollary 1.3 — Four-log theorem ····· Best rank- k

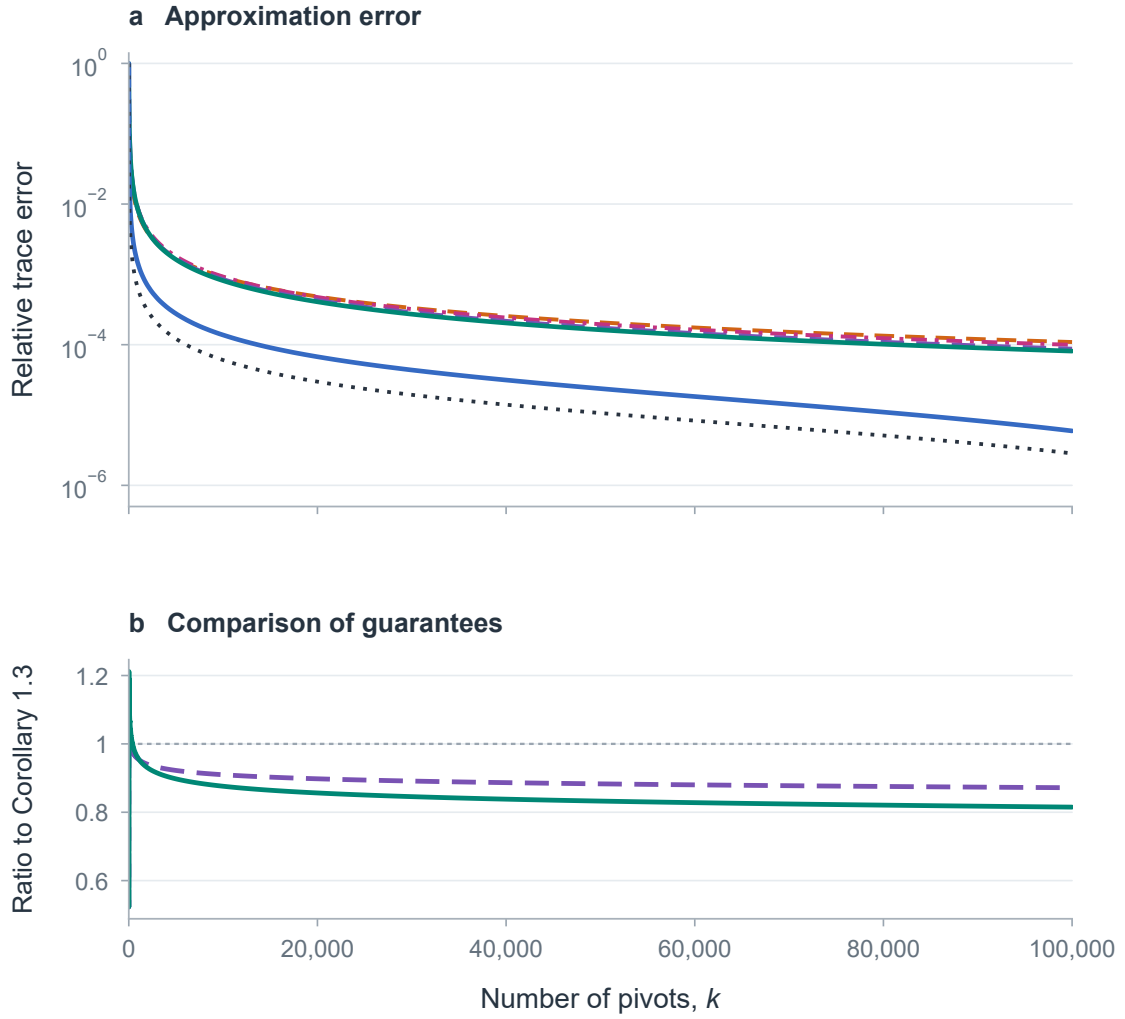


Figure 10: Brownian bridge kernel $\min(x, y) - xy$ on 131,072 equally spaced interior points, with ten original-RP trials through 100,000 pivots. Both endpoints are conditioned from the start in the interval implementation. The four-logarithm endpoint guarantee is 8.109×10^{-5} , and the empirical mean is 5.941×10^{-6} . The lower panel displays the relative improvement over Corollary 1.3.

11 Discussion

The central envelope proved here has $B_{r,\beta}/r \asymp \log \log \log r$ for fixed β , rather than a bounded value. A universal bound $\mathcal{L} \leq C^r Q^\beta$ for one fixed $\beta < 1$ would replace H by $O(r)$ and give the desired $O(r/\varepsilon)$ count through the same transfer and cooling argument. The hereditary shears control the total covariance correlation energy, but do not supply that stronger envelope. This identifies a sufficient remaining problem; it is not asserted to be necessary for the original conjecture.

A Exact structured implementation for the large kernels

Put $h = 1/(n+1)$. Between observed endpoints with indices $L < R$, let $d = R - L$ and $j = i - L \in \{1, \dots, d-1\}$. A Brownian bridge has conditional variance

$$v_j = h j(d-j)/d. \quad (52)$$

For Brownian motion, the left endpoint 0 is fixed and the rightmost unobserved interval has variance hj ; all other intervals use (52). For a Brownian bridge, both 0 and $n+1$ are fixed from the start. Prefix sums follow from the sums of the first powers and squares. These formulas follow either by a two-endpoint Schur complement or by independent Brownian increments.

For the exponential kernel put $\rho = e^{-h/0.3}$. With both endpoints observed, its conditional variance is

$$v_j = \frac{(1 - \rho^{2j})(1 - \rho^{2(d-j)})}{1 - \rho^{2d}}. \quad (53)$$

With only the left endpoint observed it is $1 - \rho^{2j}$; the right-only case is its reflection, and with neither observed it is one. Direct two-by-two conditioning proves (53). Conditional independence across an observed point follows from $k(x, z) = k(x, y)k(y, z)$ for $x \leq y \leq z$. The Brownian analog follows from independent increments. Consequently interval splitting reproduces the full Schur process at every selected coordinate.

Interval prefix traces use geometric sums and stable `expm1` evaluations. For short exponential bridges, the implementation sums the nonnegative expression (53) directly to avoid cancellation. The interval tree stores sums of child traces. At each step, draw U uniformly on $[0, T)$, where T is the total residual trace. Descend the tree by comparing U with the left-child trace, subtracting that trace when taking the right child. In the selected interval, choose the first coordinate whose prefix variance sum exceeds the remaining U . Split that interval at the pivot, compute its two child traces, and update the ancestor sums. This specifies the original coordinate RP sampling rule without constructing a dense covariance matrix.

For Brownian motion and bridge respectively, the full discrete spectra are

$$\lambda_j^{\text{BM}} = \frac{h}{4 \sin^2((2j-1)\pi/(4n+2))}, \quad \lambda_j^{\text{BB}} = \frac{h}{4 \sin^2(j\pi/(2n+2))}.$$

These follow by diagonalizing their tridiagonal inverse covariance matrices. For the exponential matrix $(\rho^{|i-j|})$, its inverse is $1/(1 - \rho^2)$ times the tridiagonal matrix with off-diagonal $-\rho$, interior diagonal $1 + \rho^2$, and endpoint diagonal 1. Its eigenvalues are reciprocals of

$$\frac{1 - 2\rho \cos \theta_j + \rho^2}{1 - \rho^2},$$

where $\theta_j \in (0, \pi)$ is the unique solution of

$$(n-1)\theta + 2 \arg(e^{i\theta} - \rho) = j\pi, \quad 1 \leq j \leq n.$$

The phase is strictly increasing: its derivative is $(n-1) + 2(1 - \rho \cos \theta)/(1 - 2\rho \cos \theta + \rho^2) > 0$. The endpoint phase values are 0 and $(n+1)\pi$. Substitution into the tridiagonal eigenvector recurrence gives the equivalent secular equation $\sin((n+1)\theta) - 2\rho \sin(n\theta) + \rho^2 \sin((n-1)\theta) = 0$. The endpoints themselves are excluded. We solve for all n roots, not a truncated spectrum.

For each of the three kernels and $n = 17, 63, 257$, every selected pivot and residual trace in a complete structured trajectory was compared with a dense Schur implementation using the same uniform variates. The maximum absolute trace discrepancy was below 4.3×10^{-14} . The full structured spectra agreed with dense eigensolves. At the large dimension, spectral traces agree with the diagonal traces to relative tolerance 2×10^{-9} . Every trial is monotone and lies above the best-rank trace error within tolerance 2×10^{-9} in normalized trace.

B Numerical protocol

Numerical calculations use Julia 1.13.0, and figures are rendered with Matplotlib 3.10.9. Each kernel has ten independent trials with the Xoshiro pseudorandom generator. For trial $j \in \{1, \dots, 10\}$, the small-kernel seeds are $2026092800 + j$ for the squared-exponential kernel and $2026092900 + j$ for Matérn-5/2. For the large kernels the seed is $202609280000 + 100i + j$, with $i = 1, 2, 3$ denoting exponential, Brownian motion, and Brownian bridge. An empirical standard error is the sample standard deviation of the ten normalized traces divided by $\sqrt{10}$.

For the small kernels, each pivot column is formed by subtracting the previous Cholesky columns from the corresponding input column. The residual diagonal is updated by subtracting the squared new column. Selected coordinates are set to zero, and negative diagonal values caused by floating-point roundoff are clipped to zero. All operations use double precision. The structured large-kernel implementation uses the conditional variance formulas in Appendix A.

The small-kernel theoretical curves are evaluated at every integer budget from 0 through 4000. For the large kernels, the theoretical grid is the sorted union of 0, the integers $1, \dots, 20$, the rounded values of 360 equally spaced points between $\log 20$ and $\log 100000$ after exponentiation, and the multiples of 500 through 100000. Every empirical integer budget is evaluated. The plotted large empirical curves retain the first 100 samples (budgets 0 through 99) and a uniformly spaced set of 2400 later samples.

At each theoretical budget k , all integer target ranks $1 \leq r \leq \min\{k, n-1\}$ are considered. For each feasible rank, the monotone scalar equation described in Section 9 determines the smallest permitted ε . The chosen root is taken from the feasible side of the bracket. The final count is checked with a 10^{-8} -pivot tolerance for floating-point inversion. Cooling exponents and killing counts were also checked with 256-bit arithmetic at every rank from 2 through 2000 and at ranks 3000, 4000, 6400, 10000, 20000, 50000, and 100000.

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